The Dependency Ordered Formal Exposition of PDEs

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Abstract

This document is a step-indexed, dependency-ordered presentation of PDE using [2]. The entries are arranged in ten levels of a topological ordering: every result appears only after all results it depends on, with circular dependencies broken at the point where the simpler formulation is introduced first. Each entry is annotated with the choice strength of its proof: ZF, ZF+DC, or ZFC.

Background — Unproved Calculus Statements

The following block gathers, in one independent list, every calculus or topology result that is taken as *given* rather than proved in this document. They are stated here as numbered background assumptions $B_1, \ldots, B_5$, before Level 0, so that the ten levels below depend only on these assumptions and on the primitives of Level 0. No proof of $B_1, \ldots, B_5$ is given in this document; their choice strength is the proof-theoretic strength of the standard proofs. Note that B_1, B_2, B_3, B_4 are ZF-level facts, whereas B_5 is a ZFC-level topology fact (Tychonoff is equivalent to the Axiom of Choice).

1	B_1 (MVT) : $\forall f \in C([a,b]) \cap D((a,b)),\ \exists c \in (a,b),\ f'(c) = (f(b) - f(a))/(b-a)$	MeanValueTheorem [5]	ZF
2	B_2 (MVTI) : $\forall w \geq 0,\ g \in C([a,b]),\ \exists c \in [a,b],\ \int_a^b w(t)g(t)\,dt = g(c)\int_a^b w(t)\,dt$	WeightedMeanValueTheoremforIntegrals	ZF
3	B_3 (IBP) : $\int_a^b u'(t)v(t)\,dt = [uv]_a^b - \int_a^b u(t)v'(t)\,dt$	IntegrationbyParts	ZF
4	B_4 (BW) : $\forall \{x_k\} \subset \mathbb{R}^n,\ \mathrm{Bdd}(x_k) \Rightarrow \exists \{x_{k_j}\},\ x \in \mathbb{R}^n,\ x_{k_j} \to x$	Bolzano – Weierstrass Theorem	ZF
5	B_5 (Tych) : $\forall \{X_i\}_{i \in I},\ (\forall i,\ \mathrm{Compact}(X_i)) \Rightarrow \mathrm{Compact}(\Pi_{i \in I} X_i)$	Tychonoff's Theorem	ZFC

Convention. All references of the form B_k in what follows point to the block above. No entry of Levels 0–9 may depend on anything outside $\{B_1, B_2, B_3, B_4, B_5\}$ plus the Level 0 primitives. The calculus results FTC, Taylor, ProductRule, ChainRule, JacobiFormula, and JacobiFormulaAdj are *not* background assumptions; they are proved inside Level 1 below.

Level 0 — Primitive Axioms and Elementary Definitions

6	ax : R – comm : $\forall (x:\mathbb{R})(y:\mathbb{R}), \mathrm{add}(x,y) = \mathrm{add}(y,x)$	AdditiveCommutativityAxiom	ZF
7	ax : R – assoc : $\forall (x:\mathbb{R})(y:\mathbb{R})(z:\mathbb{R}), \mathrm{add}(\mathrm{add}(x,y),z) = \mathrm{add}(x,\mathrm{add}(y,z))$	AdditiveAssociativityAxiom	ZF
8	ax : R – order : $\forall (x:\mathbb{R})(y:\mathbb{R})(z:\mathbb{R}), \mathrm{le}(x,y) \wedge \mathrm{le}(y,z) \Rightarrow \mathrm{le}(x,z)$	TransitivityofOrderAxiom	ZF
9	ax : R – complete : $\forall (a:\mathbb{N}\to\mathbb{R}), \mathrm{Cauchy}(a) \Rightarrow \exists (L:\mathbb{R}), \lim_{n\to\infty} a(n) = L$	CompletenessAxiom	ZF
10	ax : R – mul : $\forall (x,y,z:\mathbb{R}),\ x \leq y \wedge 0 \leq z \Rightarrow xz \leq yz$	MultiplicationCompatibility	ZF
11	ax : R – add : $\forall (x,y,z:\mathbb{R}),\ x \leq y \Rightarrow x+z \leq y+z$	AdditionCompatibility	ZF

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12	Def : M : $\mathrm{M}^{m\times n}$:= space($\mathbb{R}^{m\times n}$)	DefinitionofMatrixSpace	ZF
13	Def : S : S^{n} := space($\mathbb{R}^{n\times n}$) $\wedge$ sym(S^{n})	DefinitionofSymmetricMatrixSpace	ZF
14	Def : tr : $\forall\,(A \in \mathrm{M}^{n\times n}),\ \mathrm{tr}(A) = \sum_{i=1}^{n} A_{ii}$	DefinitionofTrace	ZF
15	Def : diag : $\mathrm{diag}(d_1,\ldots,d_n)_{ij} := d_i\delta_{ij}$	DefinitionofDiagonalMatrix	ZF

16	Def : colon : $\forall\,(A, B \in \mathrm{M}^{m\times n}),\ \mathrm{colon}(A,B) = \sum_{i,j} A_{ij}B_{ij}$	DefinitionofMatrixInnerProduct	ZF
17	Def : quad : $\forall\,(A \in \mathrm{S}^{n})(x \in \mathbb{R}^{n}),\ \mathrm{quad}(A,x) = \sum_{i,j} A_{ij}x_ix_j$	DefinitionofQuadraticForm	ZF
18	Def : sphere : $S^{n-1} := \partial B(0,1) \subset \mathbb{R}^{n}$	DefinitionofUnitSphere	ZF
19	Def : halfspace : $\mathbb{R}^{n}_{+} := \{x \in \mathbb{R}^{n} \mid x_n > 0\}$	DefinitionofUpperHalf - Space	ZF

20	Def : compactcontained : $\forall\,(U, V \subset \mathbb{R}^{n}),\ \mathrm{ccontained}(V,U) :\Leftrightarrow\ V \subset \overline{V} \subset U \wedge \mathrm{compact}(\overline{V})$	DefinitionofCompactContainment	ZF
21	Def : UT : $U_T := U \times (0,T]$	DefinitionofParabolicCylinder	ZF
22	Def : alpha : $\mathrm{alpha}(n) := \lvert B(0,1)\rvert = \frac{\pi^{n/2}}{\Gamma(n/2+1)}$	DefinitionofUnitBallVolume	ZF
23	Def : spt : $\mathrm{spt}(u) := \overline{\{x \mid u(x) \neq 0\}}$	DefinitionofSupport	ZF

24	Def : posneg : $\mathrm{pos}(u) = \max(u,0), \mathrm{neg}(u) = -\min(u,0)$	DefinitionofPositiveandNegativeParts	ZF
25	Def : sgn : $\mathrm{sgn}(x) := 1_{\{x>0\}} - 1_{\{x<0\}}$	DefinitionofSignFunction	ZF
26	Def : char : $\mathrm{char}(E,x) := 1_{\{x\in E\}}$	DefinitionofCharacteristicFunction	ZF
27	Def : Lip : $\mathrm{Lip}(u) := \sup_{x\neq y} \frac{\lvert u(x)-u(y)\rvert}{\lvert x-y\rvert}$	DefinitionofLipschitzConstant	ZF

28	Def : conv : $\mathrm{conv}(f,g)(x) := \int f(x-y)g(y)\,dy$	DefinitionofConvolution	ZF
29	Def : multi : $\forall\,(\alpha \in \mathbb{N}^{n}), \mathrm{order}(\alpha) := \alpha_1 + \cdots + \alpha_n$	DefinitionofMulti - IndexOrder	ZF
30	Def : Dalpha : $\mathrm{Da}(\alpha,u) := \partial_{x_1}^{\alpha_1}\cdots\partial_{x_n}^{\alpha_n} u$	DefinitionofMulti - IndexDerivative	ZF
31	Def : grad : $\mathrm{grad}(u) := (u_{x_1},\ldots,u_{x_n})$	DefinitionofGradient	ZF

32	Def : bigO : $\mathrm{bigO}(f,g,x_0) :\Leftrightarrow\ \exists\,C, \forall\,x\ \mathrm{near}(x,x_0), \lvert f(x)\rvert \leq C\lvert g(x)\rvert$	DefinitionofBig - ONotation	ZF
33	Def : littleo : $\mathrm{littleo}(f,g,x_0) :\Leftrightarrow\ \lim_{x\to x_0} \lvert f(x)\rvert/\lvert g(x)\rvert = 0$	DefinitionofLittle - oNotation	ZF
34	rem : grad : $Du \not\equiv \nabla u \wedge D^2u = \mathrm{Hessian} \wedge \nabla^2 u \not\equiv \mathrm{Hessian}$	RemarkonGradientNotation	ZF
35	rem : Omega : $U \not\equiv \Omega \wedge \Omega :=$ prob - space	RemarkonDomainNotation	ZF

36	Def : moll : $\mathrm{moll}(\eta) :\Leftrightarrow\ \eta(x) = C\exp\big(\frac{1}{\lvert x\rvert^2-1}\big)1_{\{\lvert x\rvert<1\}} \wedge \int \eta\,dx = 1$	DefinitionofMollifier	ZF
37	Def : Norm : $\mathrm{Norm}(\lVert\cdot\rVert) :\Leftrightarrow\ \forall\,u,v, \lVert u+v\rVert \leq \lVert u\rVert + \lVert v\rVert \wedge \forall\,\lambda, \lVert\lambda u\rVert = \lvert\lambda\rvert\lVert u\rVert \wedge \lVert u\rVert = 0 \Leftrightarrow u = 0$	DefinitionofNorm	ZF
38	Def : nu : $\mathrm{UnitNormal}(\nu,\partial U) :\Leftrightarrow\ \nu = (\nu^1,\ldots,\nu^n) \wedge \lvert\nu\rvert = 1$	DefinitionofUnitNormal	ZF
39	Def : funcCk : $C^k(U) := \{u \mid u\ \mathrm{k}-\mathrm{times}-\mathrm{contdiff}\}$	DefinitionofCkFunctions	ZF

Level 1

40	FTC : $\forall\,(f : [a,b] \to \mathbb{R}), f \in C^1 \Rightarrow \int_a^b f'(t)\,dt = f(b) - f(a)$	Fundamental Theorem of Calculus	ZF
41	Taylor : $\forall\,(f : \mathbb{R} \to \mathbb{R}), f \in C^2 \wedge x,z \in \mathbb{R} \Rightarrow \exists\,\xi \in [x,z],\ f(x) = f(z) + f'(z)(x-z) + \frac{1}{2}f''(\xi)(x-z)^2$	Taylor's Theorem with Lagrange Remainder	ZF
42	ProductRule : $\forall\,u,v \in C^1,\ (uv)' = u'v + uv'$	Product Rule for Derivatives	ZF
43	ChainRule : $\forall\,(f,g) \in C^1,\ (f\circ g)' = f'(g)\,g'$	Chain Rule for Derivatives	ZF
44	JacobiFormula : $\forall\,A \in C^1,\ (\forall\,t, \det A(t) \neq 0) \Rightarrow \frac{d}{dt}\det A = \det A \cdot \mathrm{tr}(A^{-1}A')$	Jacobi's Formula (invertible case)	ZF
45	JacobiFormulaAdj : $\forall\,A \in C^1,\ \frac{d}{dt}\det A = \mathrm{tr}\big(\mathrm{adj}(A)\cdot A'\big)$	Jacobi's Formula (adjugate form)	ZF

46	Def : normM : $\forall\,(A \in \mathrm{M}^{m\times n}), \mathrm{normM}(A) = (\mathrm{colon}(A,A))^{1/2}$	DefinitionofMatrixNorm	ZF
47	Def : ge : $\forall\,(A,B \in \mathrm{S}^{n}), \mathrm{ge}(A,B) :\Leftrightarrow$ nonnegDef(sub(A,B))	DefinitionofMatrixOrder	ZF
48	Def : parab : $\Gamma_T := \overline{U}_T \setminus U_T$	DefinitionofParabolicBoundary	ZF
49	Def : avg - ball : $\mathrm{avgB}(f,x,r) := \frac{1}{\mathrm{alpha}(n)r^n}\int_{B(x,r)} f\,dy$	DefinitionofBallAverage	ZF

50	Def : avg − sph : avgS$(f, x, r) := \frac{1}{n\ \mathrm{alpha}(n) r^{n-1}} \int_{\partial B(x,r)} f\, dS$	DefinitionofSphereAverage	ZF
51	Def : avg − set : avgE$(f, E) := \frac{1}{\mu(E)} \int_E f\, d\mu \quad (\mu(E) > 0)$	DefinitionofSetAverage [6]	ZF + DC
52	Def : Simple : Simple$(s) :\Leftrightarrow\ \exists\, E_i, a_i, s = \sum \chi_{E_i} a_i$	DefinitionofSimpleFunction	ZF
53	Def : StrongConv : StrongConv$(u, u_0) :\Leftrightarrow\ \lim_k \| u_k - u_0 \| = 0$	DefinitionofStrongConvergence	ZF

54	Def : perp : perp$(u, v) :\Leftrightarrow\ (u, v) = 0$	DefinitionofOrthogonality	ZF
55	Def : ONB : $\forall w$, ONB$(w) :\Leftrightarrow\ \big(\forall k \neq l, (w_k, w_l) = 0\big) \wedge \big(\forall k, \| w_k \| = 1\big) \wedge \big(\forall u,\ (\forall k, (u, w_k) = 0) \Rightarrow\ u = 0\big)$	DefinitionofOrthonormalBasis(withdensity)	ZF
56	Def : Sperp : Sperp$(S) := \{ u \mid \forall v \in S, (u, v) = 0 \}$	DefinitionofOrthogonalComplement	ZF
57	Def : Linear : $\forall A$, Linear$(A) :\Leftrightarrow\ \forall u, v, \lambda, \mu, A[\lambda u + \mu v] = \lambda A u + \mu A v$	DefinitionofLinearOperator	ZF

58	Def : Bdd : Bdd$(A) :\Leftrightarrow\ \exists C, \| Au \| \leq C \| u \|$	DefinitionofBoundedOperator	ZF
59	Def : ClosedOp : $\forall A$, ClosedOp$(A) :\Leftrightarrow\ u_k \to u \wedge Au_k \to v \Rightarrow\ Au = v$	DefinitionofClosedOperator	ZF
60	Def : rho : rho$(A) := \{ \eta \mid$ Bijective$(A - \eta I) \}$	DefinitionofResolventSet	ZF
61	Def : pairing : pairing$(u^*, u) := u^*(u)$	DefinitionofDualPairing	ZF

62	Def : Hess : Hess$(u) := (u_{x_i x_j})_{i,j} \in S^n$	DefinitionofHessian	ZF
63	Def : Dk : Dk$(k, u) := \{ D^\alpha u \mid |\alpha| = k \}$	Definitionofk − thOrderDerivativeSet	ZF
64	Def : dnu : dnu$(u, \nu) := \nu \cdot$ grad(u)	DefinitionofNormalDerivative	ZF
65	Def : moll − eps : moll − eps$(\eta_\epsilon, \eta) :\Leftrightarrow\ \eta_\epsilon(x) = \epsilon^{-n} \eta(x/\epsilon)$	DefinitionofStandardMollifier	ZF

66	Cauchy : $\forall a, b$, Cauchy$(a, b) :\Leftrightarrow\ ab \leq \frac{a^2}{2} + \frac{b^2}{2}$	Cauchy′ sInequality	ZF

Level 2

67	Def : InnerProduct : $\forall (\cdot,\cdot)$, InnerProduct$((\cdot,\cdot)) :\Leftrightarrow$ Sym$((\cdot,\cdot)) \wedge$ Bilinear$((\cdot,\cdot)) \wedge \forall u, (u, u) \geq 0 \wedge (u, u) = 0 \Leftrightarrow\ u = 0$	DefinitionofInnerProduct	ZF
68	Def : Lap : Lap$(u) := \Sigma_i\, u_{x_i x_i} =$ tr(Hess$(u))$	DefinitionofLaplacian	ZF
69	Def : Ckbnd : $\forall (U, k)$, Ck − bdy$(U, k) :\Leftrightarrow\ \forall x^0 \in \partial U\ \exists r > 0, \gamma \in C^k(\mathbb{R}^{n-1}) : U \cap B(x^0, r) = \{ x_n > \gamma(x_1, \ldots, x_{n-1}) \}$	DefinitionofCk − Boundary	ZF
70	Def : LebesguePoint : LebesguePoint$(x_0, f) :\Leftrightarrow\ \lim_{r \to 0}$ avgB$(|f - f(x_0)|, x_0, r) = 0$	DefinitionofLebesguePoint [6]	ZF + DC

71	Def : moll − f : moll − f$(f^\epsilon, \eta_\epsilon, f, U_\epsilon) :\Leftrightarrow\ f^\epsilon = \eta_\epsilon * f$	DefinitionofMollification [2]	ZF + DC
72	Def : Dknorm : Dknorm$(k, u) := (\Sigma_{|\alpha|=k} |D^\alpha u|^2)^{1/2}$	Definitionofk − thOrderDerivativeNorm	ZF
73	Def : StronglyMeasurable : StronglyMeasurable$(f) :\Leftrightarrow\ \exists s_k$ Simple$(s_k) \wedge s_k \to f$ a.e.	DefinitionofStrongMeasurability [6]	ZF + DC
74	Def : Cauchy : $\forall (u : \mathbb{N} \to H)$, Cauchy$(u) :\Leftrightarrow\ \forall \epsilon > 0\ \exists N, \forall k, l \geq N, \| u_k - u_l \| < \epsilon$	DefinitionofCauchySequence	ZF

75	Def : Eigenvalue : Eigenvalue$(A, \eta) :\Leftrightarrow\ N(A - \eta I) \neq \{0\}$	DefinitionofEigenvalue	ZF
76	Def : sigma : sigma$(A) := \mathbb{R} \setminus$ rho(A)	DefinitionofSpectrum	ZF
77	Def : Xstar : $X^* := \{ \phi : H \to \mathbb{R} \mid$ Linear$(\phi) \wedge$ Bdd$(\phi) \}$	DefinitionofDualSpace	ZF
78	SecondDerivCriterion : $\forall (f : \mathbb{R} \to \mathbb{R})$, $f \in C^2 \wedge (\forall x \in \mathbb{R}, f''(x) \geq 0) \Rightarrow$ Convex(f)	SecondDerivativeConvexityCriterion	ZF

79	CauchyEps : $\forall a, b, \epsilon > 0$, CauchyEps$(a, b, \epsilon) :\Leftrightarrow\ ab \leq \epsilon a^2 + \frac{b^2}{4\epsilon}$	Cauchy′ sInequalitywithEpsilon	ZF

Level 3

80	Def : SigmaAlgebra : SigmaAlgebra$(\mathcal{M}) :\Leftrightarrow\ \emptyset, \mathbb{R}^n \in \mathcal{M} \wedge \forall A \in \mathcal{M}, \mathbb{R}^n \setminus A \in \mathcal{M} \wedge \forall \{A_k\} \subset \mathcal{M}, \cup A_k, \cap A_k \in \mathcal{M}$	DefinitionofSigma − Algebra [3]	ZF
81	Def : Measurable : Measurable$(f) :\Leftrightarrow\ \forall U \subset \mathbb{R}$ Open$(U), f^{-1}(U) \in \mathcal{M}$	DefinitionofMeasurableFunction	ZF
82	Def : Summable : Summable$(f) :\Leftrightarrow\ \int |f|\, dx < \infty$	DefinitionofSummableFunction [6]	ZF + DC
83	Def : esssup : esssup$(f) := \inf\{ \mu \mid |\{ f > \mu \}| = 0 \}$	DefinitionofEssentialSupremum [6]	ZF + DC

84	Def : Banach : $\forall X$, Banach$(X) :\Leftrightarrow$ Complete$(X) \wedge$ NormedLinear(X)	DefinitionofBanachSpace [1]	ZF
85	Def : Hilbert : $\forall H$, Hilbert$(H) :\Leftrightarrow$ Banach$(H) \wedge \exists (\cdot,\cdot)$ InnerProduct$((\cdot,\cdot))$	DefinitionofHilbertSpace [1]	ZF
86	Def : Convex : $\forall (f : \mathbb{R}^n \to \mathbb{R})$, Convex$(f) :\Leftrightarrow\ \forall x, y\ \forall (\tau : [0,1]), f(\tau x + (1-\tau) y) \leq \tau f(x) + (1-\tau) f(y)$	DefinitionofConvexFunction	ZF
87	Def : Reflexive : Reflexive$(X) :\Leftrightarrow\ (X^*)^* = X$	DefinitionofReflexiveSpace [1]	ZF

88	Def : WeakConv : WeakConv$(u, u_0) :\Leftrightarrow \forall u^* \in X^*$, pairing$(u^*, u_k) \to$ pairing(u^*, u_0)	DefinitionofWeakConvergence [1]	ZF
89	Def : adjoint : adjoint$(A, A^*) :\Leftrightarrow \forall u, v, (Au, v) = (u, A^* v)$	DefinitionofAdjointOperator	ZF
90	Def : WeaklyMeasurable : WeaklyMeasurable$(f) :\Leftrightarrow \forall u^* \in X^*$, Measurable$(t \mapsto$ pairing$(u^*, f(t)))$	DefinitionofWeakMeasurability [6]	ZF + DC
91	Def : Separable : $\forall X$, Separable$(X) :\Leftrightarrow \exists D \subset X$, countable$(D) \wedge$ dense(D, X)	DefinitionofSeparableSpace [1]	ZF

92	expconvex : $\forall (x, y, \tau)$, expconvex$(x, y, \tau) :\Leftrightarrow e^{\tau x + (1-\tau)y} \leq \tau e^x + (1-\tau)e^y$	ConvexityoftheExponential	ZF
93	Young : $\forall a, b, p, q$, Young$(a, b, p, q) :\Leftrightarrow ab \leq \frac{a^p}{p} + \frac{b^q}{q}$	Young′sInequality [2]	ZF
94	CauchySchwarz : $\forall x, y$, CauchySchwarz$(x, y) :\Leftrightarrow |x \cdot y| \leq |x||y|$	Cauchy − SchwarzInequality	ZF

Level 4

95	Def : AlmostSeparablyValued : AlmostSeparablyValued$(f) :\Leftrightarrow \exists N, |N| = 0 \wedge$ Separable$(\{f(t) \mid t \notin N\})$	DefinitionofAlmostSeparablyValued [6]	ZF + DC
96	Def : Symmetric : Symmetric$(A) :\Leftrightarrow A^* = A$	DefinitionofSymmetricOperator	ZF
97	Def : CompactOp : CompactOp$(K) :\Leftrightarrow \forall \{u_k\}$ Bdd$(u_k) \Rightarrow \{Ku_k\}$ precompact	DefinitionofCompactOperator [1]	ZF
98	BanachFP : $\forall T : X \to X$, Contraction$(T) \wedge$ Complete$(X) \Rightarrow \exists! x^*, Tx^* = x^*$	Banach − PointLemma [2]	ZF

99	SeparatingHyperplane : $\forall C \subseteq \mathbb{R}^n$, Convex$(C) \Rightarrow \forall p \in \partial C \exists u \neq 0, \forall z \in C, u \cdot (z - p) \leq 0$	SupportingHyperplaneLemma [1]	ZF
100	Baire : $\forall X$, CompleteMetric$(X) \wedge \forall n$, OpenDense$(U_n, X) \Rightarrow$ Dense$(\cap_n U_n, X)$	BaireCategoryTheorem	ZF + DC
101	OpenMap : $\forall T \in \mathcal{L}(X, Y)$, Surjective$(T) \Rightarrow T$ open	OpenMappingLemma [1]	ZF + DC
102	HardyLittlewood : $\forall f \in L^1_{\text{loc}}$, avgB$(|f - f(x)|, x, r) \to 0$ a.e.	Hardy − LittlewoodLemma [6]	ZF + DC

103	BanachAlaoglu : $\forall X$, $\overline{B}_{X^*}(0, 1)$ weak* - compact	Banach − AlaogluTheorem [4]	ZFC
104	Kakutani : $\forall X$, Reflexive$(X) \Rightarrow \overline{B}_X(0, 1)$ weakly − compact	KakutaniTheorem [4]	ZFC

105	CauchySchwarzA : $\forall A, x, y$, CauchySchwarzA$(A, x, y) :\Leftrightarrow A \geq 0 \wedge A^T = A \Rightarrow |\sum a_{ij} x_i y_j| \leq (\sum a_{ij} x_i x_j)^{1/2} (\sum a_{ij} y_i y_j)^{1/2}$	Cauchy − SchwarzInequalityforPositiveMatrices	ZF

Level 5

106	YoungEps : $\forall a, b, \epsilon, p, q$, YoungEps$(a, b, \epsilon, p, q) :\Leftrightarrow ab \leq \epsilon a^p + C(\epsilon) b^q$	Young′sInequalitywithEpsilon [2]	ZF
107	Riesz : $\forall u^* \in H^* \exists! u \in H, \forall v$, pairing$(u^*, v) = (u, v)$	RieszRepresentationTheorem [1]	ZF + DC
108	BoundsSpectrum : $\forall S, m, M$, BoundsSpectrum$(S, m, M) :\Leftrightarrow$ sigma$(S) \subset [m, M] \wedge m, M \in$ sigma(S)	BoundsontheSpectrum [1]	ZF + DC
109	InverseFT : $\forall f$, Jf$(x_0) \neq 0 \Rightarrow \exists V, W$, Bijective$(f : V \to W) \wedge f^{-1} \in C^1$	InverseFunctionTheorem [2]	ZF

110	ClosedGraph : $\forall A$, ClosedOp$(A) \wedge$ Linear$(A) \Rightarrow$ Bdd(A)	ClosedGraphTheorem [1]	ZF + DC
111	GaussGreen : $\forall u$, u, $\int_U u_{x_i} = \int_{\partial U} u \nu^i \, dS \wedge \int_U$ div(u) $= \int_{\partial U}$ u $\cdot \nu \, dS$	Gauss − GreenTheorem [2]	ZF + DC
112	Coarea : $\forall u, f$, $\int_{\mathbb{R}^n} f |$grad$(u)| = \int_{-\infty}^{\infty} \left(\int_{\{u=r\}} f \, dS\right) dr$	CoareaFormula [2]	ZF + DC
113	Polar : $\forall f$, $\int_{\mathbb{R}^n} f = \int_0^\infty \left(\int_{\partial B(x_0, r)} f \, dS\right) dr$	PolarCoordinatesFormula [2]	ZF + DC

114	MonotoneConv : $\forall f_k, 0 \leq f_1 \leq f_2 \leq \cdots \Rightarrow \int \lim f_k = \lim \int f_k$	MonotoneConvergenceTheorem [6]	ZF + DC
115	Fatou : $\forall f_k$, NonnegMeasurable$(f_k) \Rightarrow \int \liminf f_k \leq \liminf \int f_k$	Fatou′sLemma [6]	ZF + DC
116	DominatedConv : $\forall f_k, f, g$, a.e. − conv$(f_k, f) \wedge |f_k| \leq g$ a.e. $\wedge$ Summable$(g) \Rightarrow \int f_k \to \int f$	DominatedConvergenceTheorem [6]	ZF + DC
117	WeakCompact : Reflexive$(X) \wedge$ Bdd$(\{u_k\}) \Rightarrow \exists \{u_{k_j}\}, u$, WeakConv$(u_{k_j}, u)$	WeakCompactnessTheorem [4]	ZFC

Level 6

118	Holder : $\forall u, v$, Holder$(u, v) :\Leftrightarrow \int_U |uv| \leq \|u\|_{L^p} \|v\|_{L^q}$	Hlder′sInequality [2]	ZF + DC
119	IntParts : $\forall u, v, \int_U u_{x_i} v = -\int_U u v_{x_i} + \int_{\partial U} uv\nu^i \, dS$	IntegrationbyParts [2]	ZF + DC
120	MovingRegions : $\forall f, U, \frac{d}{d\tau} \int_{U(\tau)} f = \int_{\partial U(\tau)} f v \cdot \nu \, dS + \int_{U(\tau)} f_\tau$	ReynoldsTransportTheorem [2]	ZF + DC
121	LebesgueDiff : $\forall f$, LocSummable$(f) \Rightarrow$ a.e.(avgB$(f, x_0, r) \to f(x_0)) \wedge$ a.e.(avgB$(|f - f(x_0)|, x_0, r) \to 0)$	LebesgueDifferentiationTheorem [6]	ZF + DC

122	LebesgueExistence : $\exists \mathcal{M}$, $\lvert\cdot\rvert$, SigmaAlgebra($\mathcal{M}$) $\wedge$ Open(U) $\wedge$ Closed(U) $\subset$ $\mathcal{M}$ $\wedge$ $\lvert B\rvert$ = vol(B) $\wedge$ countably $-$ additive($\lvert\cdot\rvert$)	ExistenceofLebesgueMeasure [6]	ZF + DC
123	Egoroff : $\forall f_k, f, A$, a.e. $-$ on(f_k, f, A) $\wedge$ $\lvert A\rvert < \infty \wedge \epsilon > 0 \Rightarrow \exists E \subset A, \lvert A \setminus E\rvert \leq \epsilon \wedge$ unif $-$ on(f_k, f, E)	Egoroff′sTheorem [6]	ZF + DC
124	CompactAdjoint : CompactOp(K) $\Rightarrow$ CompactOp(K^*)	CompactnessoftheAdjoint [4]	ZFC
125	Pettis : $\forall$ f, StronglyMeasurable(f) $\Leftrightarrow$ WeaklyMeasurable(f) $\wedge$ AlmostSeparablyValued(f)	Pettis′sMeasurabilityTheorem [6]	ZF + DC

Level 7

126	Minkowski : $\forall u, v$, Minkowski(u, v) :$\Leftrightarrow$ $\lVert u + v\rVert_{L^p} \leq \lVert u\rVert_{L^p} + \lVert v\rVert_{L^p}$	Minkowski′sInequality [2]	ZF + DC
127	GenHolder : $\forall u_k, p_k$, GenHolder(u_k, p_k) :$\Leftrightarrow$ $\int \lvert u_1 \cdots u_m\rvert \leq \Pi_k \lVert u_k\rVert_{L^{p_k}}$	GeneralizedHölder′sInequality [2]	ZF + DC
128	Interpolation : $\forall u, r, s, t, \theta$, Interpolation($u, r, s, t, \theta$) :$\Leftrightarrow$ $\lVert u\rVert_{L^r} \leq \lVert u\rVert^{\theta}_{L^s} \lVert u\rVert^{1-\theta}_{L^t}$	InterpolationInequalityforLpSpaces [2]	ZF + DC
129	MollifierProps : $\forall f, f^\epsilon, f^\epsilon \in C^\infty(U_\epsilon) \wedge f^\epsilon \to f$ a.e. $\wedge$ ($f \in C(U) \Rightarrow$ unif $-$ on $-$ compacta(f^ϵ, f)) $\wedge$ ($f \in L^p_{\text{loc}} \Rightarrow f^\epsilon \to f$ in L^p_{loc})	PropertiesofMollifiers [2]	ZF + DC

130	GreenI : $\forall u, \int_U$ Lap(u) = $\int_{\partial U}$ dnu(u, ν) dS	Green′sFirstIdentity [2]	ZF + DC
131	GreenII : $\forall u, v, \int_U$ grad(v) $\cdot$ grad(u) = $-\int_U u$Lap(v) + $\int_{\partial U} u$dnu(v, ν) dS	Green′sSecondIdentity [2]	ZF + DC
132	ImplicitFT : $\forall$ f, J_yf(x_0, y_0) $\neq 0 \Rightarrow \exists$ g $\in C^1$, f(x, g(x)) = z_0	ImplicitFunctionTheorem [2]	ZF
133	Fredholm : CompactOp(K) $\Rightarrow$ dim $N(I-K) < \infty \wedge$ ClosedOp($R(I-K)$) $\wedge$ $R(I-K)$ = Sperp($N(I-K^*)$) $\wedge$ ($N(I-K) = \{0\} \Leftrightarrow R(I-K) = H$) $\wedge$ dim $N(I-K)$ = dim $N(I-K^*)$	FredholmAlternative [1]	ZF + DC

134	Bochner : $\forall$ f, Summable(f) $\Leftrightarrow$ Summable($t \mapsto \lVert$f(t)$\rVert$) $\wedge$ $\lVert \int$f$\rVert \leq \int \lVert$f$\rVert$	Bochner′sTheorem [6]	ZF + DC
135	ArzelaAscoli : $\forall \{f_k\}$, UnifBdd(f_k) $\wedge$ Equicont(f_k) $\Rightarrow \exists \{f_{k_j}\}, f$, Continuous($f$) $\wedge$ unif $-$ on $-$ compacta(f_{k_j}, f)	Arzelà $-$ AscoliTheorem [6]	ZF + DC
136	SupportingHyperplanes : $\forall f$, Convex(f) $\Rightarrow \forall x \exists r, \forall y, f(y) \geq f(x) + r \cdot (y - x)$	SupportingHyperplaneTheorem [1]	ZF
137	LaxMilgram : $\forall a(\cdot,\cdot)$, Coercive(a) $\wedge$ Bdd(a) $\wedge$ $f \in H^* \Rightarrow \exists! u \in H, \forall v, a(u, v) = f(v)$	Lax $-$ MilgramLemma [1]	ZF + DC

Level 8

138	GreenIII : $\forall u, v, \int_U (u$Lap(v) $- v$Lap(u)) = $\int_{\partial U} (u$dnu(v, ν) $- v$dnu(u, ν)) dS	Green′sThirdIdentity [2]	ZF + DC
139	Jensen : $\forall f$, u, Convex(f) $\Rightarrow$ f(avgE(u, U)) $\leq$ avgE($f \circ$ u, U)	Jensen′sInequality [2]	ZF
140	SpectrumCompact : dim $H = \infty \wedge$ CompactOp(K) $\Rightarrow$ 0 $\in$ sigma(K) $\wedge$ sigma(K) $\setminus \{0\}$ = sigma$_p$(K) $\setminus \{0\}$ $\wedge$ finite $-$ or $-$ tends0(sigma(K) $\setminus \{0\}$)	SpectrumofaCompactOperator [1]	ZF + DC

Level 9

141	SpectralThm : Separable(H) $\wedge$ CompactOp(S) $\wedge$ Symmetric(S) $\Rightarrow \exists$ ONB $-$ of $-$ eigenvectors(S)	SpectralTheoremforCompactSymmetricOperators [1]	ZF + DC

Proofs

Level 1a — Proved calculus lemmas

Proof of Lemma 40 (FTC)

1	$f \in C^1([a, b])$	assumption	ZF
2	$F(x) := \int_a^x f'(t)\, dt$	def	ZF
3	$F(x+h) - F(x) = \int_x^{x+h} f'(t)\, dt = h \cdot f'(c), \quad c \in [x, x+h]$	Lem : 2(2)	ZF
4	$\lvert\frac{F(x+h)-F(x)}{h} - f'(x)\rvert = \lvert f'(c) - f'(x)\rvert \leq \sup_{[x,x+h]} \lvert f' - f'(x)\rvert$	Eq(3)	ZF
5	$\sup_{[x,x+h]} \lvert f' - f'(x)\rvert \to 0 \quad (h \to 0)$	assumption	ZF
6	$F'(x) = f'(x)$ for all $x \in [a, b]$	Eq(4, 5)	ZF
7	$(F - f)' = 0$ on $[a, b]$	Eq(6)	ZF
8	$\exists C \in \mathbb{R}, \forall x \in [a, b], (F - f)(x) = C$	Lem : 1(7)	ZF
9	$F(a) = 0 \Rightarrow C = -f(a)$	Eq(2, 8)	ZF
10	$F(b) = f(b) - f(a)$	Eq(8, 9)	ZF
11	$\top$	$\to$ I(1, . . . , 10)	ZF

Proof of Lemma 41 (Taylor)

12	$f \in C^2(\mathbb{R}),\ x, z \in \mathbb{R}$	assumption	ZF
13	$\int_z^x (x - t) f''(t)\, dt = \left[-(x - t) f'(t)\right]_z^x + \int_z^x f'(t)\, dt$	$Lem : 3(12)$	ZF
14	$\int_z^x (x - t) f''(t)\, dt = \int_z^x f'(t)\, dt - (x - z) f'(z)$	Eq(13)	ZF
15	$\int_z^x f'(t)\, dt = f(x) - f(z)$	$Lem : 40(12)$	ZF
16	$f(x) = f(z) + (x - z) f'(z) + \int_z^x (x - t) f''(t)\, dt$	Eq(14, 15)	ZF
17	$\exists \xi \in [x, z],\ \int_z^x (x - t) f''(t)\, dt = \frac{1}{2} f''(\xi)(x - z)^2$	$Lem : 2(12)$	ZF
18	$f(x) = f(z) + (x - z) f'(z) + \frac{1}{2} f''(\xi)(x - z)^2$	Eq(16, 17)	ZF
19	$\top$	$\to$ I(12, . . . , 18)	ZF

Proof of Lemma 42 (ProductRule)

20	$u, v \in C^1$	assumption	ZF
21	$(uv)(x + h) - (uv)(x) = u(x + h)v(x + h) - u(x)v(x)$	def	ZF
22	$= [u(x + h) - u(x)]v(x + h) + u(x)[v(x + h) - v(x)]$	Eq(21)	ZF
23	$\frac{(uv)(x+h) - (uv)(x)}{h} = \frac{u(x+h) - u(x)}{h} v(x + h) + u(x) \frac{v(x+h) - v(x)}{h}$	Eq(22)	ZF
24	$\lim_{h \to 0}(23) = u'(x)v(x) + u(x)v'(x)$	Eq(20, 23)	ZF
25	$\top$	$\to$ I(20, . . . , 24)	ZF

Proof of Lemma 43 (ChainRule)

26	$f, g \in C^1$	assumption	ZF
27	$f(g(x + h)) = f(g(x)) + f'(g(x))(g(x + h) - g(x)) + o(g(x + h) - g(x))$	def	ZF
28	$\frac{(f \circ g)(x+h) - (f \circ g)(x)}{h} = f'(g(x)) \frac{g(x+h) - g(x)}{h} + \frac{o(g(x+h) - g(x))}{h}$	Eq(27)	ZF
29	$\frac{o(g(x+h) - g(x))}{h} \to 0$	assumption	ZF
30	$\lim_{h \to 0}(28) = f'(g(x))g'(x)$	Eq(28, 29)	ZF
31	$\top$	$\to$ I(26, . . . , 30)	ZF

Proof of Lemma 44 (JacobiFormula)

32	$A \in C^1,\ \det A(t) \neq 0$	assumption	ZF
33	$\det(A + \epsilon B) = \det A + \epsilon \operatorname{tr}(\operatorname{adj}(A)B) + O(\epsilon^2)$	def	ZF
34	$\operatorname{adj}(A) = \det A \cdot A^{-1}$	def	ZF
35	$\det(A + \epsilon B) = \det A\big(1 + \epsilon \operatorname{tr}(A^{-1}B)\big) + O(\epsilon^2)$	Eq(33, 34)	ZF
36	$\frac{\det(A+\epsilon B) - \det A}{\epsilon} = \det A \cdot \operatorname{tr}(A^{-1}B) + O(\epsilon)$	Eq(35)	ZF
37	$\frac{d}{dt} \det A = \det A \cdot \operatorname{tr}(A^{-1}A')$	Eq(36)	ZF
38	$\top$	$\to$ I(32, . . . , 37)	ZF

Proof of Lemma 45 (JacobiFormulaAdj)

39	$A \in C^1$	assumption	ZF
40	$\det(A + \epsilon B) = \det A + \epsilon \operatorname{tr}(\operatorname{adj}(A)B) + O(\epsilon^2)$	def	ZF
41	$\frac{d}{dt} \det A = \operatorname{tr}(\operatorname{adj}(A)A')$	Eq(40)	ZF
42	$\top$	$\to$ I(39, . . . , 41)	ZF

Level 1 — Elementary lemmas

Proof of Lemma 66 (Cauchy)

43	$a, b \in \mathbb{R}$	assumption	ZF
44	$0 \leq (a - b)^2$	$Ax : 10$	ZF
45	$0 \leq a^2 - 2ab + b^2$	Eq(44)	ZF
46	$2ab \leq a^2 + b^2$	addMon(45)	ZF
47	$ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2$	Eq(46)	ZF
48	$\top$	$\to$ I(43, . . . , 47)	ZF

Proof of Lemma 79 (CauchyEps)

49	$a, b > 0,\ \epsilon > 0$	assumption	ZF
50	$ab = \left((2\epsilon)^{1/2}a\right)\left(b/(2\epsilon)^{1/2}\right)$	Eq	ZF
51	$ab \leq \frac{1}{2}\left((2\epsilon)^{1/2}a\right)^2 + \frac{1}{2}\left(b/(2\epsilon)^{1/2}\right)^2$	$Lem : 66(50)$	ZF
52	$ab \leq \epsilon a^2 + \frac{b^2}{4\epsilon}$	Eq(51)	ZF
53	$\top$	$\to$ I(49, . . . , 52)	ZF

Proof of Lemma 92 (expconvex)

54	$f(t) := e^t,\ f'' = e^t > 0$	def	ZF
55	$f \in C^2,\ f'' \geq 0$	Eq(54)	ZF
56	f convex	$Lem : 78(55)$	ZF
57	$e^{\tau x + (1-\tau)y} \leq \tau e^x + (1 - \tau)e^y$	$Def : 86(56)$	ZF
58	$\top$	$\to$ I(54, . . . , 57)	ZF

Proof of Lemma 93 (Young)

59	$1 < p, q < \infty,\ 1/p + 1/q = 1,\ a, b > 0$	assumption	ZF
60	$t := 1/p,\ x := \log a^p,\ y := \log b^q$	intro	ZF
61	$e^{tx+(1-t)y} \leq te^x + (1 - t)e^y$	$Lem : 92(60)$	ZF
62	$e^{(1/p)\log a^p + (1/q)\log b^q} \leq \frac{1}{p}e^{\log a^p} + \frac{1}{q}e^{\log b^q}$	Eq(61)	ZF
63	$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$	Eq(62)	ZF
64	$\top$	$\to$ I(59, . . . , 63)	ZF

Proof of Lemma 94 (CauchySchwarz)

65	$x, y \in \mathbb{R}^n$	assumption	ZF
66	$y = 0 \Rightarrow \lvert x \cdot y\rvert = 0 = \lvert x\rvert\lvert y\rvert$	Eq	ZF
67	$y \neq 0 \Rightarrow 0 \leq \lvert x - \epsilon y\rvert^2 = \lvert x\rvert^2 - 2\epsilon x \cdot y + \epsilon^2\lvert y\rvert^2$	$Ax : 10$	ZF
68	$2\epsilon x \cdot y \leq \lvert x\rvert^2 + \epsilon^2\lvert y\rvert^2$	addMon(67)	ZF
69	$x \cdot y \leq \frac{1}{2\epsilon}\lvert x\rvert^2 + \frac{\epsilon}{2}\lvert y\rvert^2$	mulMon(68)	ZF
70	$\epsilon = \lvert x\rvert/\lvert y\rvert \Rightarrow x \cdot y \leq \lvert x\rvert\lvert y\rvert$	Eq(69)	ZF
71	substitute $x \rightsquigarrow -x$ in 67–70 $\Rightarrow -x \cdot y \leq \lvert x\rvert\lvert y\rvert$	= E(67, . . . , 70)	ZF
72	$\lvert x \cdot y\rvert \leq \lvert x\rvert\lvert y\rvert$	Eq(66, 70, 71)	ZF
73	$\top$	$\to$ I(65, . . . , 72)	ZF

Proof of Lemma 106 (YoungEps)

74	$a, b > 0,\ \epsilon > 0,\ 1 < p, q < \infty,\ 1/p + 1/q = 1$	assumption	ZF
75	$ab = \big((\epsilon p)^{1/p} a\big)\big(b/(\epsilon p)^{1/p}\big)$	Eq	ZF
76	$ab \le \epsilon a^p + \frac{b^q}{q(\epsilon p)^{q/p}}$	Lem : 93(75)	ZF
77	$C(\epsilon) := (\epsilon p)^{-q/p}/q$	def	ZF
78	$ab \le \epsilon a^p + C(\epsilon) b^q$	Eq(76, 77)	ZF
79	$\top$	$\to$ I(74, . . . , 78)	ZF

Proof of Lemma 105 (CauchySchwarzA)

80	$A \succeq 0,\ A^T = A$	assumption	ZF
81	$(x, y) \mapsto x^T A y$ is an inner product	Def : 67(80)	ZF
82	$\lvert x^T A y\rvert \le (x^T A x)^{1/2} (y^T A y)^{1/2}$	Lem : 94(81)	ZF
83	$\top$	$\to$ I(80, . . . , 82)	ZF

Proof of Lemma 78 (SecondDerivCriterion)

84	$f \in C^2,\ \forall x,\ f''(x) \ge 0$	assumption	ZF
85	$x, y \in \mathbb{R},\ t \in [0, 1],\ z := (1 - t)x + ty$	intro	ZF
86	$\exists \xi_1,\ f(x) = f(z) + f'(z)(x - z) + \frac{1}{2} f''(\xi_1)(x - z)^2$	Lem : 41(84, 85)	ZF
87	$f(x) \ge f(z) + f'(z)(x - z)$	Eq(86, 84)	ZF
88	$\exists \xi_2,\ f(y) = f(z) + f'(z)(y - z) + \frac{1}{2} f''(\xi_2)(y - z)^2$	Lem : 41(84, 85)	ZF
89	$f(y) \ge f(z) + f'(z)(y - z)$	Eq(88, 84)	ZF
90	$(1 - t)f(x) + tf(y) \ge (1 - t)(f(z) + f'(z)(x - z)) + t(f(z) + f'(z)(y - z)) = f(z)$	addMon(87, 89)	ZF
91	$f(\tau x + (1 - \tau)y) \le \tau f(x) + (1 - \tau) f(y)$	Def : 86(85, 90)	ZF
92	$\top$	$\to$ I(84, . . . , 91)	ZF

Level 2 — Foundational theorems

Proof of Lemma 98 (BanachFP)

93	T contraction, X complete	assumption	ZF
94	$\exists \theta < 1,\ \forall x, y,\ d(Tx, Ty) \le \theta d(x, y)$	def	ZF
95	$x_{n+1} := T(x_n)$	def	ZF
96	$d(x_{n+1}, x_n) \le \theta^n d(x_1, x_0)$	Eq(94, 95)	ZF
97	$d(x_n, x_m) \le \frac{\theta^n}{1-\theta} d(x_1, x_0)$ for $m > n$	Eq(96)	ZF
98	(x_n) is Cauchy	Def : 74(97)	ZF
99	$\exists x^*,\ x_n \to x^*$	Ax : 9(98)	ZF
100	$Tx^* = x^*$	Eq(95, 99)	ZF
101	uniqueness	Eq(94)	ZF
102	$\top$	$\to$ I(93, . . . , 101)	ZF

Proof of Lemma 99 (SeparatingHyperplane)

103	$C \subset \mathbb{R}^n$ convex, $p \in \partial C$	assumption	ZF
104	$\exists p_k \notin \bar{C},\ p_k \to p$	def	ZF
105	$\exists \pi_k \in \bar{C},\ \lVert p_k - \pi_k\rVert = \mathrm{dist}(p_k, C)$	$\exists$E(104)	ZF
106	$\forall z \in C,\ (p_k - \pi_k) \cdot (z - \pi_k) \le 0$	Def : 86(103, 105)	ZF
107	$u_k := (p_k - \pi_k)/\lVert p_k - \pi_k\rVert,\ \lVert u_k\rVert = 1$	def	ZF
108	$\exists \{u_{k_j}\},\ u_{k_j} \to u,\ \lVert u\rVert = 1$	Lem : 4(107)	ZF
109	$\forall z \in C,\ u \cdot (z - p) \le 0$	Eq(106, 107, 108)	ZF
110	$\top$	$\to$ I(103, . . . , 109)	ZF

Proof of Lemma 100 (Baire)

111	X complete metric, U_n open dense	assumption	ZF + DC
112	$\bar{B}(x_0, r_0) \subset U_1$	def	ZF + DC
113	$\exists x_n, r_n,\ \bar{B}(x_n, r_n) \subset U_n \cap B(x_{n-1}, r_{n-1}),\ r_n \to 0$	$\exists$E(112)	ZF + DC
114	(x_n) Cauchy $\Rightarrow\ \exists x^*$	$Ax:9(113)$	ZF + DC
115	$x^* \in \cap_n U_n$	Eq(113, 114)	ZF + DC
116	$\top$	$\to$ I(111, . . . , 115)	ZF + DC

Proof of Lemma 101 (OpenMap)

117	$T \in \mathcal{L}(X, Y)$ surjective	assumption	ZF + DC
118	$X = \cup_n \bar{B}(0, n),\ Y = T(X) = \cup_n T(\bar{B}(0, n))$	Eq(117)	ZF + DC
119	$\exists n,\ \overline{T(B(0, n))}$ has nonempty interior	$Lem:100(118)$	ZF + DC
120	$\exists r > 0,\ \bar{B}_Y(0, r) \subseteq T(\bar{B}(0, n))$	$\exists$E(119)	ZF + DC
121	$T(B(0, 1)) \supseteq B_Y(0, r/n)$	Eq(120)	ZF + DC
122	T open	def	ZF + DC
123	$\top$	$\to$ I(117, . . . , 122)	ZF + DC

Proof of Theorem 104 (Kakutani)

124	X reflexive	assumption	ZFC
125	$\bar{B}_X(0, 1) \subseteq X = (X^*)^*$	$Def:87(124)$	ZFC
126	$\bar{B}_{X^{**}}(0, 1)$ weak*-compact	$Thm:103$	ZFC
127	$\bar{B}_X(0, 1)$ weakly compact	Eq(125, 126)	ZFC
128	$\top$	$\to$ I(124, . . . , 127)	ZFC

Proof of Lemma 102 (HardyLittlewood)

129	$f \in L^1_{\text{loc}}$	assumption	ZF + DC
130	$M(f)(x) := \sup_{r>0}$ avgB$(\lvert f\rvert, x, r)$	$Def:49(129)$	ZF + DC
131	$\lVert \text{WeakL}^1(M(f))\rVert \le C\lVert f\rVert_{L^1}$	def	ZF + DC
132	avgB$(\lvert f - f(x)\rvert, x, r) \to 0$ a.e.	Eq(129, 131)	ZF + DC
133	$\top$	$\to$ I(129, . . . , 132)	ZF + DC

Proof of Theorem 103 (BanachAlaoglu)

134	X Banach	assumption	ZFC
135	$\bar{B}_{X^*}(0, 1) \subset \prod_{x \in X}[-\lVert x\rVert, \lVert x\rVert]$ via $\varphi \mapsto (\varphi(x))_x$	def	ZFC
136	$\prod_x [-\lVert x\rVert, \lVert x\rVert]$ compact	$Lem:5$	ZFC
137	$\bar{B}_{X^*}(0, 1)$ closed in the product	def	ZFC
138	$\bar{B}_{X^*}(0, 1)$ weak*-compact	Eq(136, 137)	ZFC
139	$\top$	$\to$ I(134, . . . , 138)	ZFC

Level 3 — Measure and integration

Proof of Theorem 122 (LebesgueExistence)

140	$\mathrm{Box}(I) := \prod(a_i, b_i]$, $\mathrm{vol}(I) = \prod(b_i - a_i)$	def	ZF + DC
141	$\mathrm{outmeas}(A) := \inf\{\sum \mathrm{vol}(I_k) : A \subseteq \cup I_k\}$	def	ZF + DC
142	outmeas monotone, countably subadditive, $\mathrm{outmeas}(\emptyset) = 0$	Eq(141)	ZF + DC
143	$\mathcal{M} := \{A : \forall E, \mathrm{outmeas}(E) = \mathrm{outmeas}(E \cap A) + \mathrm{outmeas}(E \setminus A)\}$	def	ZF + DC
144	$\mathcal{M}$ is $\sigma-$ algebra	$Def : 80(143)$	ZF + DC
145	$\lvert\cdot\rvert := \mathrm{outmeas}\lvert_{\mathcal{M}}$, countably additive	def	ZF + DC
146	$\forall I,\ \lvert I\rvert = \mathrm{vol}(I) \Rightarrow \lvert B\rvert = \mathrm{vol}(B)$	Eq(141)	ZF + DC
147	$\mathrm{open}(U) \Rightarrow U \in \mathcal{M}$	Eq(143)	ZF + DC
148	$\mathrm{closed}(F) \Rightarrow F \in \mathcal{M}$	Eq(144, 147)	ZF + DC
149	$\top$	$\to$ I(140, . . . , 148)	ZF + DC

Proof of Theorem 114 (MonotoneConv)

150	$0 \le f_1 \le f_2 \le \cdots$, $f := \sup_k f_k$	assumption	ZF + DC
151	$\int f_k \uparrow L$ with $L \le \int f$	def	ZF + DC
152	$0 \le s \le f$, simple s, $E_k := \{f_k \ge (1-\epsilon)s\} \uparrow X$	$Def : 52(150)$	ZF + DC
153	$(1-\epsilon)\int_{E_k} s \le \int f_k \le L \Rightarrow \int s \le L$	Eq(151, 152)	ZF + DC
154	$\int f \le L$	$\forall$ E(153)	ZF + DC
155	$\int f = L = \lim \int f_k$	Eq(151, 154)	ZF + DC
156	$\top$	$\to$ I(150, . . . , 155)	ZF + DC

Proof of Theorem 115 (Fatou)

157	$f_k \ge 0$ measurable	assumption	ZF + DC
158	$g_k := \inf_{j \ge k} f_j$, g_k nondecreasing	def	ZF + DC
159	$\int g_k \uparrow \int \liminf f_k$	$Thm : 114(158)$	ZF + DC
160	$\int g_k \le \inf_{j \ge k} \int f_j$	Eq(157, 158)	ZF + DC
161	$\int \liminf f_k \le \liminf \int f_k$	Eq(159, 160)	ZF + DC
162	$\top$	$\to$ I(157, . . . , 161)	ZF + DC

Proof of Theorem 116 (DominatedConv)

163	$\lvert f_k\rvert \le g$, $g \in L^1$, $f_k \to f$ a.e.	assumption	ZF + DC
164	$\int(g + f) \le \liminf \int(g + f_k)$	$Thm : 115(163)$	ZF + DC
165	$\int f \le \liminf \int f_k$	Eq(164)	ZF + DC
166	$\int(g - f) \le \liminf \int(g - f_k)$	$Thm : 115(163)$	ZF + DC
167	$\limsup \int f_k \le \int f$	Eq(166)	ZF + DC
168	$\int f \le \liminf \int f_k \le \limsup \int f_k \le \int f$	Eq(165, 167)	ZF + DC
169	$\int f_k \to \int f$	Eq(168)	ZF + DC
170	$\top$	$\to$ I(163, . . . , 169)	ZF + DC

Proof of Theorem 118 (Holder)

171	$\lVert u\rVert_{L^p} = \lVert v\rVert_{L^q} = 1$	assumption	ZF + DC
172	$\lvert u(x)v(x)\rvert \le \lvert u(x)\rvert^p/p + \lvert v(x)\rvert^q/q$ a.e.	$Lem : 93(171)$	ZF + DC
173	$\int\lvert uv\rvert \le \lVert u\rVert_{L^p}^p/p + \lVert v\rVert_{L^q}^q/q = 1$	Eq(172)	ZF + DC
174	$\int\lvert uv\rvert \le \lVert u\rVert_{L^p}\lVert v\rVert_{L^q}$	Eq(173, 171)	ZF + DC
175	$\top$	$\to$ I(171, . . . , 174)	ZF + DC

Proof of Theorem 126 (Minkowski)

176	$u, v \in L^p$	assumption	ZF + DC
177	$\|u+v\|_{Lp}^p = \int \|u+v\|^p \le \int \|u+v\|^{p-1}(\|u\|+\|v\|)$	Eq	ZF + DC
178	$\int \|u+v\|^{p-1}\|u\| \le \|u+v\|_{Lp}^{p-1}\|u\|_{Lp}$	Thm : 118(177)	ZF + DC
179	$\int \|u+v\|^{p-1}\|v\| \le \|u+v\|_{Lp}^{p-1}\|v\|_{Lp}$	Thm : 118(177)	ZF + DC
180	$\|u+v\|_{Lp}^p \le \|u+v\|_{Lp}^{p-1}(\|u\|_{Lp}+\|v\|_{Lp})$	addMon(178, 179)	ZF + DC
181	$\|u+v\|_{Lp} \le \|u\|_{Lp}+\|v\|_{Lp}$	Eq(180)	ZF + DC
182	$\top$	$\to$ I(176, . . . , 181)	ZF + DC

Proof of Theorem 127 (GenHolder)

183	$1 \le p_1, \ldots, p_m \le \infty$, $\sum 1/p_k = 1$	assumption	ZF + DC
184	$m = 2 \Rightarrow$ GenHolder	Thm : 118(183)	ZF + DC
185	IH : $m-1$ case holds, $q_j := p_j/(1-1/p_1)$	assumption	ZF + DC
186	$\int \|u_1 \cdots u_m\| \le \|u_1\|_{Lp_1}\|u_2 \cdots u_m\|_{L1/(1-1/p_1)}$	Thm : 118(185)	ZF + DC
187	$\|u_2 \cdots u_m\|_{L1/(1-1/p_1)} \le \Pi_{k\ge 2}\|u_k\|_{Lp_k}$	Eq(185)	ZF + DC
188	$\int \|u_1 \cdots u_m\| \le \Pi \|u_k\|_{Lp_k}$	Eq(186, 187)	ZF + DC
189	$\top$	$\to$ I(183, . . . , 188)	ZF + DC

Proof of Theorem 128 (Interpolation)

190	$1/r = \theta/s + (1-\theta)/t$	assumption	ZF + DC
191	r/s and r/t conjugate exponents	def	ZF + DC
192	$\int \|u\|^r = \int \|u\|^{\theta r}\|u\|^{(1-\theta)r}$	Eq	ZF + DC
193	$\le (\int \|u\|^s)^{\theta r/s}(\int \|u\|^t)^{(1-\theta)r/t}$	Thm : 118(192)	ZF + DC
194	$\|u\|_{Lr} \le \|u\|_{Ls}^{\theta}\|u\|_{Lt}^{1-\theta}$	Eq(193)	ZF + DC
195	$\top$	$\to$ I(190, . . . , 194)	ZF + DC

Proof of Theorem 121 (LebesgueDiff)

196	$f \in L^1_{\mathrm{loc}}$, $u(x) :=$ avgB($\|f-f(x)\|, x, r$)	Def : 49	ZF + DC
197	u subharmonic − like, $\int_{B(x,r)} u \to 0$	Lem : 102(196)	ZF + DC
198	avgB($\|f-f(x_0)\|, x_0, r$) $\to 0$ a.e. x_0	$\forall$E(197)	ZF + DC
199	avgB(f, x_0, r) $\to f(x_0)$ a.e.	Eq(198)	ZF + DC
200	$\top$	$\to$ I(196, . . . , 199)	ZF + DC

Proof of Theorem 123 (Egoroff)

201	$f_k \to f$ a.e. on A, $\|A\| < \infty$	assumption	ZF + DC
202	$A_j^m := \cap_{k\ge j}\{\|f_k - f\| < 1/m\}$	def	ZF + DC
203	$A_j^m \uparrow$ $\bar{A}^m$, $\bar{A}^m$ is a.e. equal to A	Eq(201, 202)	ZF + DC
204	$\|A \setminus A_j^m\| \to 0$	Eq(203)	ZF + DC
205	$\exists j(m)$, $\|A \setminus A_{j(m)}^m\| < \epsilon/2^m$	$\exists$E(204)	ZF + DC
206	$E := \cap_m A_{j(m)}^m$	def	ZF + DC
207	$\|A \setminus E\| \le \epsilon$, unif conv. on E	Eq(205, 206)	ZF + DC
208	$\top$	$\to$ I(201, . . . , 207)	ZF + DC

Level 4 — Hilbert and spectral

Proof of Theorem 107 (Riesz)

209	$u^* \in H^*$	assumption	ZF + DC
210	$N(u^*) := \{v : u^*(v) = 0\}$ closed subspace	def	ZF + DC
211	$u^* = 0 \Rightarrow u = 0$	Eq	ZF + DC
212	$u^* \neq 0 \Rightarrow \exists w \in N(u^*)^\perp,\ w \neq 0,\ \|w\| = 1$	$Def : 56(210)$	ZF + DC
213	$u := \overline{u^*(w)}w$	def	ZF + DC
214	$\forall v,\ u^*(v) = (u, v)$	$Def : 54(212, 213)$	ZF + DC
215	uniqueness	Eq(214)	ZF + DC
216	$\top$	$\to$ I(209, . . . , 215)	ZF + DC

Proof of Lemma 137 (LaxMilgram)

217	a coercive, bounded, $f \in H^*$	assumption	ZF + DC
218	$\exists! A \in \mathcal{L}(H),\ a(u, v) = (Au, v)$	$Thm : 107(217)$	ZF + DC
219	$\|Au\| \geq c\|u\|$	Eq(217, 218)	ZF + DC
220	A injective, bounded	Eq(219)	ZF + DC
221	$R(A)$ closed, A surjective	Eq(217, 220)	ZF + DC
222	$\exists! u,\ Au = h,\ h =$ Riesz(f)	$\exists$E(221)	ZF + DC
223	$\top$	$\to$ I(217, . . . , 222)	ZF + DC

Proof of Lemma 108 (BoundsSpectrum)

224	S bounded self adjoint, $m := \inf \sigma(S),\ M := \sup \sigma(S)$	assumption	ZF + DC
225	$\eta > M \Rightarrow (\eta I - S)$ coercive $\Rightarrow$ bijective	$Lem : 137(224)$	ZF + DC
226	$\eta < m \Rightarrow (S - \eta I)$ coercive $\Rightarrow$ bijective	$Lem : 137(224)$	ZF + DC
227	$\sigma(S) \subset [m, M]$	$\forall$E(225, 226)	ZF + DC
228	$\|Mu - Su\| \leq C(Mu - Su, u)^{1/2}$	$Lem : 94(227)$	ZF + DC
229	$\|u_k\| = 1,\ (Su_k, u_k) \to M \Rightarrow \|Mu_k - Su_k\| \to 0$	Eq(228)	ZF + DC
230	$M \in \sigma(S)$	RAA(229)	ZF + DC
231	$m \in \sigma(S)$	Eq(224, 230)	ZF + DC
232	$\top$	$\to$ I(224, . . . , 231)	ZF + DC

Proof of Theorem 133 (Fredholm)

233	K compact	assumption	ZF + DC
234	$\dim N(I - K) = \infty \Rightarrow \exists$ ONB $(u_k) \subset N(I - K),\ Ku_k = u_k,\ \|u_k - u_l\| = \sqrt{2}$	$Def : 55(233)$	ZF + DC
235	$\|Ku_k - Ku_l\| = \sqrt{2} \Rightarrow \neg$ compact(K)	$\perp$ I(234)	ZF + DC
236	$\dim N(I - K) < \infty$	RAA(234, 235)	ZF + DC
237	$\exists \gamma > 0,\ \|u - Ku\| \geq \gamma\|u\|$ on $N(I - K)^\perp$	Eq(236)	ZF + DC
238	$R(I - K)$ closed	Eq(237)	ZF + DC
239	$R(I - K) = N(I - K^*)^\perp$	$Def : 56(238)$	ZF + DC
240	$N(I - K) = 0 \Leftrightarrow R(I - K) = H$	Eq(239)	ZF + DC
241	$\dim N(I - K) = \dim N(I - K^*)$	$\forall$E(239)	ZF + DC
242	$\top$	$\to$ I(233, . . . , 241)	ZF + DC

Proof of Theorem 140 (SpectrumCompact)

243	K compact, $\dim H = \infty$	assumption	ZF + DC
244	$0 \notin \sigma(K) \Rightarrow\ K$ invertible $\Rightarrow\ I = K \circ K^{-1}$ compact	$Def:76(243)$	ZF + DC
245	$\dim H = \infty\ \Rightarrow\ \neg$ compact(I)	$\bot$ I(244)	ZF + DC
246	$0 \in \sigma(K)$	RAA(244, 245)	ZF + DC
247	$\eta \in \sigma(K) \setminus \{0\}\ \Rightarrow\ \eta$ eigenvalue	$Thm:133(246)$	ZF + DC
248	$\eta_k \to \eta \neq 0 \Rightarrow\ \neg$ compact(K)	$\bot$ I(247)	ZF + DC
249	$\sigma(K) \setminus \{0\}$ tends to 0	Eq(248)	ZF + DC
250	$\top$	$\to$ I(243, ..., 249)	ZF + DC

Proof of Theorem 141 (SpectralThm)

251	H separable, S compact symmetric	assumption	ZF + DC
252	$H_k := N(S - \eta_k I)$, $\dim H_k < \infty$	$Thm:133(251)$	ZF + DC
253	$H_k \perp H_l$ for $k \neq l$	$Def:96(252)$	ZF + DC
254	$\tilde{H} := \overline{\oplus H_k}$, $S(\tilde{H}^\perp) \subset \tilde{H}^\perp$	$Def:56(253)$	ZF + DC
255	$\tilde{S} := S\|_{\tilde{H}^\perp}$ compact symmetric, $\sigma(\tilde{S}) = \{0\}$	def	ZF + DC
256	$(\tilde{S}u, u) = 0\ \forall u \Rightarrow\ \tilde{S} = 0$	$Lem:108(255)$	ZF + DC
257	$\tilde{H}^\perp \subset N(S) \subset \tilde{H} \Rightarrow\ \tilde{H} = H$	$\forall$E(256)	ZF + DC
258	S has an ONB of eigenvectors	$\forall$I(252, 257)	ZF + DC
259	$\top$	$\to$ I(251, ..., 258)	ZF + DC

Level 5 — Multivariate calculus and identities

Proof of Theorem 109 (InverseFT)

260	$WLOG\ x_0 = 0$, $f(0) = 0$, $Df(0) = I$	def	ZF
261	$f(x) = x + R(x)$, $x \mapsto R(x)$ has DR small	def	ZF
262	$\|R(x) - R(y)\| \leq \frac{1}{2}\|x - y\| \Rightarrow\ \|f(x) - f(y)\| \geq \frac{1}{2}\|x - y\|$	$Lem:1(261)$	ZF
263	$\forall z$ near 0, $\exists x$, $f(x) = z$	$Lem:98(261)$	ZF
264	$g := f^{-1}$ exists locally	def	ZF
265	$Dg(z) = Df(g(z))^{-1}$	$Lem:43(264)$	ZF
266	$f \in C^k \Rightarrow\ g \in C^k$	$\forall$I(265)	ZF
267	$\top$	$\to$ I(260, ..., 266)	ZF

Proof of Theorem 132 (ImplicitFT)

268	$f(x_0, y_0) = z_0$, $D_y f(x_0, y_0)$ invertible	assumption	ZF
269	$F(x, y) := (f(x, y), x) : U \to \mathbb{R}^{m+n}$	def	ZF
270	$DF = \begin{pmatrix} D_x f & D_y f \\ I & 0 \end{pmatrix}$, $\det DF = \pm \det D_y f \neq 0$	def	ZF
271	F locally invertible	$Thm:109(270)$	ZF
272	$g(x) :=$ 2nd$(F^{-1}(z_0, x))$	def	ZF
273	$f(x, g(x)) = z_0$	Eq(271, 272)	ZF
274	$f \in C^k \Rightarrow\ g \in C^k$	$\forall$E(266, 271)	ZF
275	$\top$	$\to$ I(268, ..., 274)	ZF

Proof of Theorem 110 (ClosedGraph)

276	A closed linear, X, Y Banach	assumption	ZF + DC
277	$G(A) := \{(u, Au)\} \subset X \times Y$ closed	def	ZF + DC
278	$X \times Y$ Banach, $\|(u,v)\| = \|u\| + \|v\|$	Def : 84(276)	ZF + DC
279	$P : G(A) \to X$, $(u, Au) \mapsto u$ bijective linear	def	ZF + DC
280	P^{-1} bounded	Lem : 101(279)	ZF + DC
281	$\|Au\| \leq \|P^{-1}\| \|u\|$	Eq(280)	ZF + DC
282	A bounded	Def : 58(281)	ZF + DC
283	$\top$	$\to$ I(276, . . . , 282)	ZF + DC

Proof of Theorem 111 (GaussGreen)

284	U smooth domain, $u \in C^1(\bar{U})$	assumption	ZF + DC
285	$\{\zeta_k\}$ partition of unity subordinate to U	def	ZF + DC
286	$y = \Phi(x)$ flattens boundary, $\det D\Phi = 1$	Def : 69(284)	ZF + DC
287	$\int_{y_n>0} \partial_{y_n}(\zeta u)\, dy = \int_{y_n=0} \zeta u\, dy'$	Lem : 40(286)	ZF + DC
288	$\int_U (\zeta u)_{x_n}\, dx = \int_{\partial U} \zeta u\, \nu^n\, dS$	def	ZF + DC
289	$\sum_k \zeta_k = 1 \Rightarrow \int_U u_{x_n} = \int_{\partial U} u\nu^n\, dS$	Eq(288)	ZF + DC
290	$\int_U u_{x_i} = \int_{\partial U} u\nu^i\, dS$	$\forall$E(289)	ZF + DC
291	$\int_U \operatorname{div} \mathbf{u} = \int_{\partial U} \mathbf{u} \cdot \nu\, dS$	Eq(290)	ZF + DC
292	$\top$	$\to$ I(284, . . . , 291)	ZF + DC

Proof of Theorem 119 (IntParts)

293	$u, v \in C^1(\bar{U})$	assumption	ZF + DC
294	$\int_U (uv)_{x_i} = \int_{\partial U} uv\nu^i\, dS$	Thm : 111(293)	ZF + DC
295	$(uv)_{x_i} = u_{x_i} v + uv_{x_i}$	Lem : 42(293)	ZF + DC
296	$\int_U u_{x_i} v = -\int_U uv_{x_i} + \int_{\partial U} uv\nu^i\, dS$	Eq(294, 295)	ZF + DC
297	$\top$	$\to$ I(293, . . . , 296)	ZF + DC

Proof of Theorem 130 (GreenI)

298	$u \in C^2(\bar{U})$	assumption	ZF + DC
299	$\int_U u_{x_i x_i} = \int_{\partial U} u_{x_i} \nu^i\, dS$	Thm : 119(298)	ZF + DC
300	$\int_U \Delta u = \int_{\partial U} \partial u/\partial\nu\, dS$	Def : 68(299)	ZF + DC
301	$\top$	$\to$ I(298, . . . , 300)	ZF + DC

Proof of Theorem 131 (GreenII)

302	$u, v \in C^2(\bar{U})$	assumption	ZF + DC
303	$\int_U u_{x_i} v_{x_i} = -\int_U uv_{x_i x_i} + \int_{\partial U} uv_{x_i} \nu^i\, dS$	Thm : 119(302)	ZF + DC
304	$\int_U \nabla v \cdot \nabla u = -\int_U u\Delta v + \int_{\partial U} u\partial v/\partial\nu\, dS$	Def : 31(303)	ZF + DC
305	$\top$	$\to$ I(302, . . . , 304)	ZF + DC

Proof of Theorem 138 (GreenIII)

306	$u, v \in C^2(\bar{U})$	assumption	ZF + DC
307	$\int_U \nabla u \cdot \nabla v = -\int_U v\Delta u + \int_{\partial U} v\partial u/\partial\nu$	Thm : 131(306)	ZF + DC
308	$\int_U (u\Delta v - v\Delta u) = \int_{\partial U} (u\partial v/\partial\nu - v\partial u/\partial\nu)$	Eq(304, 307)	ZF + DC
309	$\top$	$\to$ I(306, . . . , 308)	ZF + DC

Proof of Theorem 129 (MollifierProps)

310	$f \in L^1_{\mathrm{loc}},\ f^\epsilon = \eta_\epsilon * f$	$Def : 71$	ZF + DC
311	$D^\alpha f^\epsilon(x) = \int D^\alpha \eta_\epsilon(x - y) f(y)\, dy$	def	ZF + DC
312	$f^\epsilon \in C^\infty$	$\forall$ I(311)	ZF + DC
313	$\lvert f^\epsilon(x) - f(x)\rvert \le C\mathrm{avgB}(\lvert f - f(x)\rvert, x, \epsilon) \to 0$	$Lem : 102$(310)	ZF + DC
314	$f^\epsilon \to f$ a.e.	$\forall$ E(313)	ZF + DC
315	$f \in C(U) \Rightarrow \sup_V \lvert f^\epsilon - f\rvert \le C\omega(\epsilon) \to 0$	Eq(313, 314)	ZF + DC
316	$\lVert f^\epsilon - f\rVert_{L^p(V)} \le 2\lVert f - g\rVert_{L^p(W)} + \lVert g^\epsilon - g\rVert_{L^p(V)}$	$Thm : 118$(310)	ZF + DC
317	$f^\epsilon \to f$ in L^p_{loc}	Eq(315, 316)	ZF + DC
318	$\top$	$\to$ I(310, . . . , 317)	ZF + DC

Level 6 — Consequences

Proof of Theorem 112 (Coarea)

319	u Lipschitz, f measurable	assumption	ZF + DC
320	ApproxSimple(f)	assumption	ZF + DC
321	$\int f\lvert\nabla u\rvert = \int(\int_{\{u=r\}} f\, dS)\, dr$	Eq(319, 320)	ZF + DC
322	$\top$	$\to$ I(319, . . . , 321)	ZF + DC

Proof of Theorem 113 (Polar)

323	$WLOG\ x_0 = 0,\ u(x) := \lvert x\rvert$, Lipschitz($u$), $\lvert\nabla u\rvert = 1$ a.e.	def	ZF + DC
324	$\int f\lvert\nabla u\rvert = \int_0^\infty (\int_{\partial B(0,r)} f\, dS)\, dr$	$Thm : 112$(323)	ZF + DC
325	$\int f = \int_0^\infty (\int_{\partial B(0,r)} f\, dS)\, dr$	Eq(324)	ZF + DC
326	$\top$	$\to$ I(323, . . . , 325)	ZF + DC

Proof of Theorem 120 (MovingRegions)

327	$X(\tau, \cdot) : U(0) \to U(\tau),\ \partial_\tau X = v$	def	ZF + DC
328	$\int_{U(\tau)} f = \int_{U(0)} f(X(\tau, y), \tau) J\, dy,\ J := \det \partial_y X$	def	ZF + DC
329	$\partial_\tau J = J\, \mathrm{div}_x(v)$	$Lem : 44$(328)	ZF + DC
330	$\partial_\tau[fJ] = [f_\tau + \nabla f \cdot v]J + fJ\, \mathrm{div}(v)$	$Lem : 42$(329)	ZF + DC
331	$\frac{d}{d\tau}\int_{U(\tau)} f = \int_{U(\tau)}[f_\tau + \mathrm{div}(fv)]$	Eq(328, 330)	ZF + DC
332	$\int_{U(\tau)} \mathrm{div}(fv) = \int_{\partial U(\tau)} fv \cdot \nu$	$Thm : 111$	ZF + DC
333	$\frac{d}{d\tau}\int_{U(\tau)} f = \int_{\partial U(\tau)} fv \cdot \nu + \int_{U(\tau)} f_\tau$	Eq(331, 332)	ZF + DC
334	$\top$	$\to$ I(327, . . . , 333)	ZF + DC

Proof of Theorem 139 (Jensen)

335	f convex, U open bounded, u summable	assumption	ZF
336	$\forall p,\ \exists r,\ \forall q,\ f(q) \ge f(p) + r \cdot (q - p)$	$Thm : 136$(335)	ZF
337	$p := \mathrm{avgE}(u, U),\ q := u(x)$	def	ZF
338	$\int_U f(u) \ge \lvert U\rvert f(p) + r \cdot (\int_U u - \lvert U\rvert p) = \lvert U\rvert f(p)$	$Def : 51$(337)	ZF
339	$f(\mathrm{avgE}(u, U)) \le \mathrm{avgE}(f \circ u, U)$	Eq(338)	ZF
340	$\top$	$\to$ I(335, . . . , 339)	ZF

Proof of Theorem 135 (ArzelaAscoli)

341	$\{f_k\}$ unif. bdd, equicont.	assumption	ZF + DC
342	$WLOG\ K = \bar{B}(0,R)$	def	ZF + DC
343	$\exists\, \epsilon-$ net $\{x_1,\ldots,x_N\} \subset K$	$\exists$ E(342)	ZF + DC
344	$\exists\{k_j\},\ \forall i,\ f_{k_j}(x_i)$ converges	$\exists$ E(343)	ZF + DC
345	$\lvert f_{k_j}(x) - f_{k_l}(x)\rvert \le 2\epsilon + \lvert f_{k_j}(x_i) - f_{k_l}(x_i)\rvert$	$\forall$ E(344)	ZF + DC
346	$f_{k_j} \to f$ unif.	Eq(344, 345)	ZF + DC
347	$\top$	$\to$ I(341, ..., 346)	ZF + DC

Proof of Theorem 136 (SupportingHyperplanes)

348	$f : \mathbb{R}^n \to \mathbb{R}$ convex, $x \in \mathbb{R}^n$	assumption	ZF
349	epi$(f) := \{(z,t) : t \ge f(z)\}$	def	ZF
350	epi(f) convex	Def : 86(348, 349)	ZF
351	$(x, f(x)) \in \partial$epi(f)	def	ZF
352	$\exists\,(r,-\alpha) \neq 0,\ (r,-\alpha)\cdot((z,t) - (x,f(x))) \le 0$	Lem : 99(350, 351)	ZF
353	$\alpha \ge 0$	Eq(352)	ZF
354	$\alpha > 0$	RAA(353, 352)	ZF
355	$f(z) \ge f(x) + r\cdot(z-x)$	Eq(352, 354)	ZF
356	$\forall x,\ \exists r,\ \forall y,\ f(y) \ge f(x) + r\cdot(y-x)$	$\forall$I(355)	ZF
357	$\top$	$\to$ I(348, ..., 356)	ZF

Proof of Theorem 124 (CompactAdjoint)

358	K compact	assumption	ZFC
359	$\{u_k\}$ bdd $\Rightarrow\ \exists\{u_{k_j}\},\ u_{k_j} \rightharpoonup u$	Thm : 117(358)	ZFC
360	$\lVert K^* u_{k_j} - K^* u\rVert^2 = (KK^*(u_{k_j} - u), u_{k_j} - u)$	Def : 89(359)	ZFC
361	$KK^*(u_{k_j} - u) \to 0$	Def : 97(359)	ZFC
362	$\lVert K^*(u_{k_j} - u)\rVert^2 \to 0$	Eq(360, 361)	ZFC
363	K^* compact	Def : 97(362)	ZFC
364	$\top$	$\to$ I(358, ..., 363)	ZFC

Proof of Theorem 117 (WeakCompact)

365	X reflexive, $\{u_k\}$ bdd	assumption	ZFC
366	$\bar{B}(0,M)$ weakly compact	Thm : 104(365)	ZFC
367	$\exists\{u_{k_j}\} \subset \{u_k\},\ u \in X,\ u_{k_j} \rightharpoonup u$	$\exists$ E(366)	ZFC
368	$\top$	$\to$ I(365, ..., 367)	ZFC

Proof of Theorem 134 (Bochner)

369	f strongly measurable	assumption	ZF + DC
370	$\exists\{s_k\}$ simple, $s_k \to f$	Def : 73(369)	ZF + DC
371	$\int \lVert s_k' - f\rVert \to 0$	Thm : 116(370)	ZF + DC
372	f summable $\Leftrightarrow$ $\lVert f\rVert$ summable	Eq(371)	ZF + DC
373	$\lVert \int f\rVert \le \int \lVert f\rVert$	Eq(371, 372)	ZF + DC
374	$\top$	$\to$ I(369, ..., 373)	ZF + DC

Proof of Theorem 125 (Pettis)

375	f strongly measurable ⇒ weakly measurable	$Def: 73$	ZF + DC
376	f strongly measurable ⇒ almost sep. valued	$\forall$ E(375)	ZF + DC
377	weakly meas. + almost sep. valued ⇒ strongly meas.	$\wedge$ I(375, 376)	ZF + DC
378	$\top$	$\rightarrow$ I(375, . . . , 377)	ZF + DC

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